

Homework 3

1. Determine the convergence and absolute convergence of the following series:

$$(1) \sum_{n=2}^{\infty} \frac{i^n}{\ln n};$$

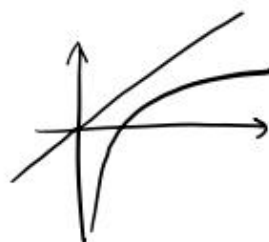
$$(2) \sum_{n=1}^{\infty} \frac{i^n}{n}.$$

1. 判断下列级数的收敛性与绝对收敛性

$$(1) \sum_{n=2}^{\infty} \frac{i^n}{\ln n}$$

$$\left| \frac{i^n}{\ln n} \right| = \frac{1}{\ln n} > \frac{1}{n}$$

$$\sum_{n=2}^{\infty} \frac{1}{n} \quad \text{即 } u_n = \frac{1}{n}$$



$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \right| = 1$$

$$\frac{u_n}{u_{n+1}} = 1 + \frac{\lambda}{n} + O(n^{-2}) = \frac{n+1}{n} = 1 + \frac{1}{n}$$

($\lambda = a + ib$)

$$a=1 \Rightarrow \sum_{n=2}^{\infty} \frac{1}{n} \text{ 发散 (Gauss 判别法)}$$

$$\Rightarrow \sum_{n=2}^{\infty} \left| \frac{i^n}{\ln n} \right| \text{ 发散}$$

$$\sum_{n=2}^{\infty} \frac{i^n}{\ln n} = \frac{-1}{\ln 2} + \frac{-i}{\ln 3} + \frac{1}{\ln 4} + \frac{i}{\ln 5} + \dots$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{\ln 2n} + i \sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(2n+1)}$$

收敛 收敛

$$\lim_{n \rightarrow \infty} \frac{1}{\ln 2n} = 0, \quad \frac{1}{\ln 2n} > \frac{1}{\ln 2(n+1)} \quad (n \rightarrow \infty)$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{\ln 2n} \text{ 收敛}$$

$$\text{同理, } \sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(2n+1)} \text{ 收敛}$$

$$\Rightarrow \sum_{n=2}^{\infty} \frac{i^n}{\ln n} \text{ 收敛}$$

$$\text{综上: } \sum_{n=2}^{\infty} \frac{i^n}{\ln n} \text{ 收敛, 但不绝对收敛}$$

莱布尼兹判别法:

① 交错级数: $\sum_{n=1}^{\infty} (-1)^n u_n$, $u_n > 0$

② $\lim_{n \rightarrow \infty} u_n = 0$, $u_n \geq u_{n+1}$
 $\Rightarrow$ 收敛

$$(2) \sum_{n=1}^{\infty} \frac{i^n}{n}$$

$$\left| \frac{i^n}{n} \right| = \frac{1}{n}, \quad \sum_{n=1}^{\infty} \frac{1}{n} \text{ 发散} \Rightarrow \text{不绝对收敛}$$

$$\sum_{n=1}^{\infty} \frac{i^n}{n} = \frac{i}{1} + \frac{-1}{2} + \frac{-i}{3} + \frac{1}{4} + \dots$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} + i \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$$

$$\lim_{n \rightarrow \infty} \frac{1}{2n} = 0, \quad \frac{1}{2n} > \frac{1}{2(n+1)} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \text{ 收敛}$$

$$\text{同理 } \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \text{ 收敛}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{i^n}{n} \text{ 收敛, 但不绝对收敛}$$

2. Try to identify the convergence regions for the following series:

$$(1) \sum_{n=1}^{\infty} z^{n!};$$

$$(2) \sum_{n=1}^{\infty} \left(\frac{z}{1+z} \right)^n;$$

$$(3) \sum_{n=1}^{\infty} (-1)^n (z^2 + 2z + 2)^n;$$

$$(4) \sum_{n=1}^{\infty} 2^n \sin \frac{z}{3^n}.$$

$$(1) \sum_{n=1}^{\infty} z^{n!}$$

$$\text{当 } \lim_{n \rightarrow \infty} \left| \frac{z^{(n+1)!}}{z^{n!}} \right| = \lim_{n \rightarrow \infty} |z^{n+1}| < 1 \text{ 时, 级数绝对收敛}$$

$$\text{当 } \lim_{n \rightarrow \infty} |z^{n+1}| > 1 \text{ 时, 级数发散}$$

$$\Rightarrow \text{收敛区域: } |z| < 1$$

$$(2) \sum_{n=1}^{\infty} \left(\frac{z}{1+z} \right)^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{\left(\frac{z}{1+z} \right)^{n+1}}{\left(\frac{z}{1+z} \right)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{z}{1+z} \right| \Rightarrow \text{收敛区域为 } \left| \frac{z}{1+z} \right| < 1$$

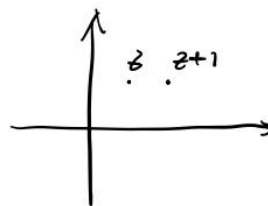
$$\text{令 } z = x + iy$$

$$\left| \frac{z}{z+1} \right| = \frac{|z|}{|z+1|} = \frac{x^2+y^2}{(x+1)^2+y^2} < 1$$

$$\Rightarrow x^2+y^2 < (x+1)^2+y^2$$

$$\Rightarrow x > -\frac{1}{2}$$

收敛区域: $\text{Re}[z] > -\frac{1}{2}$



$$(3) \sum_{n=1}^{\infty} (-1)^n (z^2 + 2z + 2)^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (z^2 + 2z + 2)^{n+1}}{(-1)^n (z^2 + 2z + 2)^n} \right| = \lim_{n \rightarrow \infty} |z^2 + 2z + 2| < 1$$

$$\Rightarrow \text{收敛区域: } |z^2 + 2z + 2| < 1$$

$$(4) \sum_{n=1}^{\infty} 2^n \sin \frac{z}{3^n}$$

$$\lim_{n \rightarrow \infty} \left| \frac{2^{n+1} \sin \frac{z}{3^{n+1}}}{2^n \sin \frac{z}{3^n}} \right| = \lim_{n \rightarrow \infty} \left| 2 \cdot \frac{\sin \left(\frac{1}{3} \cdot \frac{z}{3^n} \right)}{\sin \frac{z}{3^n}} \right|$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$\text{取 } \lim_{n \rightarrow \infty} \left| 2 \cdot \frac{\sin \frac{z}{3^{n+1}}}{\sin \frac{z}{3^n}} \right| = \lim_{n \rightarrow \infty} \left| 2 \cdot \frac{\sin \frac{z}{3^{n+1}}}{3\sin \frac{z}{3^{n+1}} - 4\sin^3 \frac{z}{3^{n+1}}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{2}{3 - 4\sin^2 \frac{z}{3^{n+1}}} \right| < 1$$

$\Rightarrow$ 全平面收敛

3. Try to find the convergence radius of the following military ranks:

$$(1) \sum_{n=1}^{\infty} \frac{1}{n^n} z^n;$$

$$(2) \sum_{n=1}^{\infty} \frac{1}{2^n n^n} z^n;$$

$$(3) \sum_{n=1}^{\infty} \frac{n!}{n^n} z^n;$$

$$(4) \sum_{n=1}^{\infty} \frac{(-)^n}{2^{2n} (n!)^2} z^n;$$

$$(5) \sum_{n=1}^{\infty} n^{\ln n} z^n;$$

$$(6) \sum_{n=1}^{\infty} \frac{1}{2^{2n}} z^{2n};$$

$$(7) \sum_{n=1}^{\infty} \frac{\ln n^n}{n!} z^n;$$

$$(8) \sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^n z^n.$$

$$(1) \sum_{n=1}^{\infty} \frac{1}{n^n} z^n$$

$$\lim_{n \rightarrow \infty} |n^n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} n \Rightarrow R = \infty$$

$$R = \lim_{n \rightarrow \infty} \left| \frac{1}{c_n} \right|^{\frac{1}{n}}$$

$$(2) \sum_{n=1}^{\infty} \frac{1}{2^n n^n} z^n$$

$$\lim_{n \rightarrow \infty} |2^n n^n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} 2n \Rightarrow R = \infty$$

$$(3) \sum_{n=1}^{\infty} \frac{n!}{n^n} z^n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{\frac{n!}{n^n}}{\frac{(n+1)!}{(n+1)^{n+1}}} \right| &= \lim_{n \rightarrow \infty} \left| \left(\frac{n+1}{n} \right)^n \right| = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n \\ &= \lim_{n \rightarrow \infty} e^{n \ln \left(1 + \frac{1}{n} \right)} = \lim_{n \rightarrow \infty} e^{n \cdot \frac{1}{n}} = e \end{aligned}$$

$$\begin{array}{l} x \rightarrow 0 \\ \ln(1+x) \sim x \end{array}$$

$$\Rightarrow R = e$$

$$(4) \sum_{n=1}^{\infty} \frac{(-1)^n}{2^{2n} (n!)^2} z^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^n}{2^{2n} (n!)^2}}{\frac{(-1)^{n+1}}{2^{2(n+1)} [(n+1)!]^2}} \right| = \lim_{n \rightarrow \infty} \frac{2^{2(n+1)} [(n+1)!]^2}{2^{2n} (n!)^2} = \lim_{n \rightarrow \infty} 4(n+1)^2 = \infty$$

$$\Rightarrow R = \infty$$

$$(5) \sum_{n=1}^{\infty} n^{\ln n} z^n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} |n^{\ln n}|^{\frac{1}{n}} &= \lim_{n \rightarrow \infty} n^{\frac{\ln n}{n}} = \lim_{n \rightarrow \infty} e^{\frac{\ln n}{n} \ln n} = \lim_{n \rightarrow \infty} e^{\frac{(\ln n)^2}{n}} \\ &= \lim_{n \rightarrow \infty} e^{\frac{2 \ln n}{n}} = \lim_{n \rightarrow \infty} e^{\frac{2}{n}} = 1 \end{aligned}$$

$$\Rightarrow R = \frac{1}{1} = 1$$

$$(6) \sum_{n=1}^{\infty} \frac{1}{2^{2n}} z^{2n}$$

$$\sum_{n=1}^{\infty} \frac{z^{2n}}{2^{2n}} = \sum_{n=1}^{\infty} \left(\frac{z}{2}\right)^{2n}$$

$$\lim_{n \rightarrow \infty} \left| \left(\frac{z}{2}\right)^{2n} \right|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{z}{2}\right)^2 < 1$$

$$\Rightarrow |z| < 2, \quad R = 2$$

$$(7) \sum_{n=1}^{\infty} \frac{\ln n^n}{n!} z^n$$

$$\sum_{n=1}^{\infty} \frac{\ln n^n}{n!} z^n = \sum_{n=1}^{\infty} \frac{n \ln n}{n!} z^n = \sum_{n=1}^{\infty} \frac{\ln n}{(n-1)!} z^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{\ln n}{(n-1)!}}{\frac{\ln(n+1)}{n!}} \right| = \lim_{n \rightarrow \infty} \frac{n \ln n}{\ln(n+1)} = \lim_{n \rightarrow \infty} \frac{\ln n + 1}{\frac{1}{n+1}} = \lim_{n \rightarrow \infty} (n+1)(1 + \ln n) = \infty$$

$$\Rightarrow R = \infty$$

$$(8) \sum_{n=1}^{\infty} \left(1 - \frac{1}{n}\right)^n z^n$$

$$\lim_{n \rightarrow \infty} \left| \left(1 - \frac{1}{n}\right)^n \right|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1$$

$$\Rightarrow R = \frac{1}{1} = 1$$

Chapter 5

4. Expand the following function into a Taylor series at the specified point and give its radius of convergence:

(1) $1 - z^2$, expand at $z = 1$;

(2) $\sin z$, expand at $z = n\pi$;

(3) $\frac{1}{1+z+z^2}$, expand at $z = 0$;

(4) $\frac{\sin z}{1-z}$, expand at $z = 0$;

(5) $\exp\{\frac{1}{1-z}\}$, expand at $z = 0$; (You can only ask for the first four)

Problem 4

将下列函数在指定点展开为 Taylor 级数，并给出其收敛半径：

(1) $1 - z^2$, 在 $z = 1$ 展开；

$$1 - z^2 = -(z-1)(z+1) = -(z-1)[2 + (z-1)] = -2(z-1) - (z-1)^2$$

$$|z| < \infty$$

(2) $\sin z$, 在 $z = n\pi$ 展开；

$$(\sin z)_{z=n\pi}^{(k)} = \sin\left(n\pi + \frac{k}{2}\pi\right) = \begin{cases} 0, k = 2m \\ (-1)^{n+m}, k = 2m+1 \end{cases}$$

$$\sin z = \sum_{k=0}^{\infty} \frac{1}{k!} (\sin z)_{z=n\pi}^{(k)} (z - n\pi)^k = \sum_{m=0}^{\infty} \frac{(-1)^{n+m}}{(2m+1)!} (z - n\pi)^{2m+1}$$

$$|z| < \infty$$

(3) $\frac{1}{1+z+z^2}$, 在 $z = 0$ 展开；

$$\begin{aligned}
\frac{1}{1+z+z^2} &= \frac{1}{(z-e^{i2\pi/3})(z-e^{-i2\pi/3})} = \frac{1}{\sqrt{3}i} \left(\frac{1}{z-e^{i2\pi/3}} - \frac{1}{z-e^{-i2\pi/3}} \right) \\
&= \frac{1}{\sqrt{3}i} \left(\frac{e^{i2\pi/3}}{1-e^{i2\pi/3}z} - \frac{e^{-i2\pi/3}}{1-e^{-i2\pi/3}z} \right) \\
&= \frac{1}{\sqrt{3}i} \sum_{n=0}^{\infty} \left[e^{i2\pi/3} (e^{i2\pi/3}z)^n - e^{-i2\pi/3} (e^{-i2\pi/3}z)^n \right] \\
&= \frac{1}{\sqrt{3}i} \sum_{n=0}^{\infty} \left[e^{i2(n+1)\pi/3} - e^{-i2(n+1)\pi/3} \right] z^n = \frac{2}{\sqrt{3}} \sum_{n=0}^{\infty} \sin \left[\frac{2}{3}(n+1)\pi \right] z^n \quad (|z| < 1)
\end{aligned}$$

(4) $\frac{\sin z}{1-z}$, 在 $z=0$ 展开;

$$\left(\frac{\sin z}{1-z} \right)_{z=0}^{(n)} = \sum_{k=0}^n \frac{n!}{k!(n-k)!} \left(\frac{1}{1-z} \right)_{z=0}^{(n-k)} (\sin z)_{z=0}^{(k)} = \sum_{k=0}^n \frac{n!}{k!} \sin \frac{k}{2}\pi = \sum_{m=0}^{[(n-1)/2]} \frac{(-1)^m n!}{(2m+1)!}$$

$$\frac{\sin z}{1-z} = \left(\frac{\sin z}{1-z} \right)_{z=0} + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{\sin z}{1-z} \right)_{z=0}^{(n)} z^n = \sum_{n=1}^{\infty} \sum_{m=0}^{[(n-1)/2]} \frac{(-1)^m}{(2m+1)!} z^n, \quad (|z| < 1)$$

(5) $\exp \left\{ \frac{1}{1-z} \right\}$, 在 $z=0$ 展开 (可只求前四项).

$$\text{记 } f(z) = e^{\frac{1}{1-z}}, \text{ 则 } f(0) = e. \quad f'(z) = \frac{1}{(1-z)^2} e^{\frac{1}{1-z}}, \quad f'(0) = e.$$

$$f''(z) = \left[\frac{2}{(1-z)^3} + \frac{1}{(1-z)^4} \right] e^{\frac{1}{1-z}}, \quad f''(0) = 3e.$$

$$f'''(z) = \left[\frac{6}{(1-z)^4} + \frac{6}{(1-z)^5} + \frac{1}{(1-z)^6} \right] e^{\frac{1}{1-z}}, \quad f'''(0) = 13e.$$

$$f^{(4)}(z) = \left[\frac{24}{(1-z)^5} + \frac{36}{(1-z)^6} + \frac{12}{(1-z)^7} + \frac{1}{(1-z)^8} \right] e^{\frac{1}{1-z}}, \quad f^{(4)}(0) = 73e.$$

$$\begin{aligned}
f(z) &= f(0) + f'(0)z + \frac{1}{2}f''(0)z^2 + \frac{1}{6}f'''(0)z^3 + \frac{1}{24}f^{(4)}(0)z^4 + \cdots \\
&= e \left(1 + z + \frac{3}{2}z^2 + \frac{13}{6}z^3 + \frac{73}{24}z^4 + \cdots \right) \quad (|z| < 1)
\end{aligned}$$

5. Find the sum of the following infinite series, taking care to give the corresponding convergence region:

$$(1) \sum_{n=0}^{\infty} \frac{1}{2n+1} z^{2n+1};$$

$$(2) \sum_{n=0}^{\infty} \frac{1}{(2n)!} z^{2n};$$

$$(3) \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(n+m)!}{n! m!} \left(\frac{z}{2}\right)^{n+m};$$

$$(4) \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{(n+m+p)!}{n! m! p!} \left(\frac{z}{3}\right)^{n+m+p}.$$

Problem 5 / Problem 6 (Chapter 5 3)

$$(1) \sum_{n=0}^{\infty} \frac{1}{2n+1} z^{2n+1};$$

记 $f(z) = \sum_{n=0}^{\infty} \frac{1}{2n+1} z^{2n+1}$, 则 $f'(z) = \sum_{n=0}^{\infty} z^{2n} = \frac{1}{1-z^2}$, ($|z| < 1$)

$f(z) = f(0) + \int_0^z \frac{1}{1-t^2} dt = \frac{1}{2} \int_0^z \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt = \frac{1}{2} \ln \frac{1+z}{1-z}$, 为使 $f(0) = 0$, 规定

$$\ln \frac{1+z}{1-z} \Big|_{z=0} = 0.$$

$$(2) \sum_{n=0}^{\infty} \frac{1}{(2n)!} z^{2n};$$

记 $f(z) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} z^{2n}$, 则 $f'(z) = \sum_{n=1}^{\infty} \frac{1}{(2n-1)!} z^{2n-1}$, $f''(z) = \sum_{n=1}^{\infty} \frac{1}{(2n-2)!} z^{2n-2} = \sum_{k=0}^{\infty} \frac{1}{(2k)!} z^{2k}$,

所以有 $f''(z) - f(z) = 0$, 由初始条件 $f(0) = 1$, $f'(0) = 0$ 解此方程得 $f(z) = \cosh z$.

$$|z| < \infty$$

$$(3) \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(n+m)!}{n! m!} \left(\frac{z}{2}\right)^{n+m};$$

令 $n+m = k$, 则:

$$\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(n+m)!}{n! m!} \left(\frac{z}{2}\right)^{n+m} = \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} C_m^k \left(\frac{z}{2}\right)^k = \sum_{k=0}^{\infty} 2^k \left(\frac{z}{2}\right)^k = \frac{1}{1-z}$$

$$|z| < 2, \operatorname{Re}[z] < 1 \text{ 的公共区域}$$

$$(4) \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{(n+m+p)!}{n! m! p!} \left(\frac{z}{3}\right)^{n+m+p}.$$

令 $n+m+p=k$ ，则：

$$\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{(n+m+p)!}{n! m! p!} \left(\frac{z}{3}\right)^{n+m+p} = \sum_{k=0}^{\infty} \left(\sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{k!}{(k-n-m)! m! p!} \right) \left(\frac{z}{3}\right)^k = \sum_{k=0}^{\infty} z^k = \frac{1}{1-z}$$

$|z-2|<1$, $|z|<3$, $\operatorname{Re}[z]<\frac{3}{2}$ 的公共区域

6. Find the Laurent expansion of the following function: (Chapter 5 4)

(1) $\frac{1}{z^2(z-1)}$, expand around $z=1$;

(2) $\frac{1}{z^2(z-1)}$, the expansion area is $1<|z|<\infty$;

(3) $\frac{1}{z^2-3z+2}$, the expansion area is $1<|z|<2$;

(4) $\frac{1}{z^2-3z+2}$, the expansion area is $2<|z|<\infty$;

(5) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, the expansion area is $3<|z|<4$;

(6) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, the expansion area is $4<|z|<\infty$;

Problem 6 / Problem 7 (Chapter 5 4)

求下列函数的 Laurent 展开：

(1) $\frac{1}{z^2(z-1)}$, 在 $z=1$ 附近展开；

$$\frac{1}{z^2} = [1+(z-1)]^{-2} = \sum_{n=0}^{\infty} \frac{(-2)(-2-1)(-2-2)\cdots(-2-n+1)}{n!} (z-1)^n$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (n+1)!}{n!} (z-1)^n = \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^n \quad (|z-1|<1)$$

$$\frac{1}{z^2(z-1)} = (z-1)^{-1} \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^n = \sum_{k=-1}^{\infty} (-1)^{k+1} (k+2) (z-1)^k, \quad (0<|z-1|<1)$$

(2) $\frac{1}{z^2(z-1)}$, 展开区域为 $1 < |z| < \infty$;

$$\frac{1}{z^2(z-1)} = \frac{1}{z^3(1-\frac{1}{z})} = \frac{1}{z^3} \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^n = \sum_{n=-\infty}^0 z^{n-3}, |z| > 1$$

(3) $\frac{1}{z^2-3z+2}$, 展开区域为 $1 < |z| < 2$;

$$\begin{aligned} \frac{1}{z^2-3z+2} &= \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2} \frac{1}{1-z/2} - z^{-1} \frac{1}{1-z^{-1}} \\ &= -\frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} z^n - z^{-1} \sum_{n=0}^{\infty} z^{-n} = -\sum_{n=-1}^{\infty} z^n - \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} z^n \quad (1 < |z| < 2) \end{aligned}$$

(4) $\frac{1}{z^2-3z+2}$, 展开区域为 $2 < |z| < \infty$;

$$\frac{1}{z^2-3z+2} = z^{-1} \frac{1}{1-2z^{-1}} - z^{-1} \frac{1}{1-z^{-1}} = \sum_{k=-2}^{\infty} (2^{-k-1} - 1) z^k \quad (2 < |z| < \infty)$$

(5) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, 展开区域为 $3 < |z| < 4$;

$$\begin{aligned} \frac{(z-1)(z-2)}{(z-3)(z-4)} &= 1 + \frac{6}{z-4} - \frac{2}{z-3} = 1 - \frac{3}{2} \frac{1}{1-z/4} - 2z^{-1} \frac{1}{1-3z^{-1}} \\ &= 1 - \frac{3}{2} \sum_{n=0}^{\infty} 4^{-n} z^n - 2 \sum_{n=-1}^{\infty} 3^{-n-1} z^n \quad (3 < |z| < 4) \end{aligned}$$

(6) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, 展开区域为 $4 < |z| < \infty$.

$$\begin{aligned} \frac{(z-1)(z-2)}{(z-3)(z-4)} &= 1 + 6z^{-1} \frac{1}{1-4z^{-1}} - 2z^{-1} \frac{1}{1-3z^{-1}} \\ &= 1 + \sum_{n=-1}^{\infty} \left(\frac{3}{2} \cdot 4^{-n} - \frac{2}{3} \cdot 3^{-n} \right) z^n \quad (4 < |z| < \infty) \end{aligned}$$

7. Determine the nature of the isolated singularity of the following functions, and in the case of the poles, determine its order: (Chapter 5 6)

$$(1) \frac{1}{z^2 + a^2}, a \neq 0;$$

$$(2) \frac{\cos az}{z^2};$$

$$(3) \frac{\cos az - \cos bz}{z^2}, a^2 \neq b^2;$$

$$(4) \frac{\sin z}{z^2} - \frac{1}{z};$$

$$(5) \cos \frac{1}{\sqrt{z}};$$

$$(6) \frac{\sqrt{z}}{\sin \sqrt{z}};$$

$$(7) \frac{1}{(z-1)\ln z};$$

$$(8) \int_0^z \frac{\sinh \sqrt{\zeta}}{\sqrt{\zeta}} d\zeta.$$

Problem 7 / Problem 8 (Chapter 5 6)

$$(1) \frac{1}{z^2 + a^2}, a \neq 0;$$

$\pm ai$ 是一阶极点。 $f\left(\frac{1}{t}\right) = \frac{t^2}{1+a^2t^2}$, 0 不是 $f(1/t)$ 的奇点, 故 ∞ 不是 $f(z)$ 的奇点。

$$(2) \frac{\cos az}{z^2};$$

$$0 \text{ 是二阶极点; } f\left(\frac{1}{t}\right) = t^2 \left[1 - \frac{1}{2!} \left(\frac{a}{t}\right)^2 + \frac{1}{4!} \left(\frac{a}{t}\right)^4 - + \dots \right] = t^2 - \frac{a^2}{2!} + \frac{a^4}{4!} t^{-2} - + \dots$$

0 是 $f(1/t)$ 的本性奇点, 所以 ∞ 是 $f(z)$ 的本性奇点。

$$(3) \frac{\cos az - \cos bz}{z^2}, a^2 \neq b^2;$$

$$\begin{aligned} f(z) &= z^{-2} \left[\left(1 - \frac{1}{2!} a^2 z^2 + \frac{1}{4!} a^4 z^4 - + \dots \right) - \left(1 - \frac{1}{2!} b^2 z^2 + \frac{1}{4!} b^4 z^4 - + \dots \right) \right] \\ &= -\frac{1}{2!} (a^2 - b^2) + \frac{1}{4!} (a^4 - b^4) z^2 - + \dots \end{aligned}$$

所以 0 是 $f(z)$ 的可去奇点。 $f(1/t) = -\frac{1}{2!} (a^2 - b^2) + \frac{1}{4!} (a^4 - b^4) t^{-2} - + \dots$, ∞ 是 $f(z)$ 的本性奇点。

$$(4) \frac{\sin z}{z^2} - \frac{1}{z};$$

$$f(z) = z^{-2} \left(z - \frac{1}{3!} z^3 + \frac{1}{5!} z^5 - + \dots \right) - z^{-1} = -\frac{1}{3!} z + \frac{1}{5!} z^3 - + \dots, \quad 0 \text{ 是可去奇点。}$$

$$f\left(\frac{1}{t}\right) = -\frac{1}{3!} t^{-1} + \frac{1}{5!} t^{-3} - + \dots, \quad \infty \text{ 是 } f(z) \text{ 的本性奇点。}$$

$$(5) \cos \frac{1}{\sqrt{z}};$$

$$\cos \frac{1}{\sqrt{z}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (z)^{-n}, \quad z=0 \text{ 为本性极点}$$

$$(6) \frac{\sqrt{z}}{\sin \sqrt{z}};$$

$$\lim_{z \rightarrow 0} f(z) = 1, \text{ 所以 } 0 \text{ 是可去奇点。}$$

$$\lim_{z \rightarrow n^2 \pi^2} (z - n^2 \pi^2) f(z) = \lim_{z \rightarrow n^2 \pi^2} \frac{(z - n^2 \pi^2) \sqrt{z}}{\sin \sqrt{z}} = \lim_{z \rightarrow n^2 \pi^2} \frac{\sqrt{z} + \frac{(z - n^2 \pi^2)}{2\sqrt{z}}}{\frac{\cos \sqrt{z}}{2\sqrt{z}}} = (-1)^n 2n^2 \pi^2$$

所以 $n^2 \pi^2$ ($n=1, 2, 3, \dots$) 是一阶极点。由于 $n^2 \pi^2 \rightarrow \infty$, 所以 ∞ 点的任意邻域包含无穷个奇点, 即 ∞ 点为非孤立奇点。

$$(7) \frac{1}{(z-1) \ln z};$$

$$\lim_{z \rightarrow 1} \frac{(z-1)^2}{(z-1) \ln z} = \lim_{z \rightarrow 1} \frac{z-1}{\ln z} = \lim_{z \rightarrow 1} z = 1, \quad z=1 \text{ 为二阶极点}$$

$$(8) \int_0^z \frac{\sinh \sqrt{\zeta}}{\sqrt{\zeta}} d\zeta.$$

$$\int_0^z \frac{\sinh \sqrt{\xi}}{\sqrt{\xi}} d\xi = \left(2 \cosh \sqrt{\xi} \right) \Big|_{\xi=0}^{\xi=z} = 2 \cosh \sqrt{z} - 2, \quad z = \infty \text{ 为本性极点}$$